For V2 presented at the start of the workshop see.
https://drive.google.com/file/d/1FMtYYjT06w0LhKgRvqTO7tQqemmYN1Er/view?usp=sharing

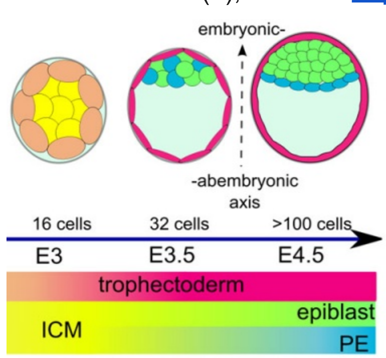
This version will potentially evolve. Please add comments where clarification is needed

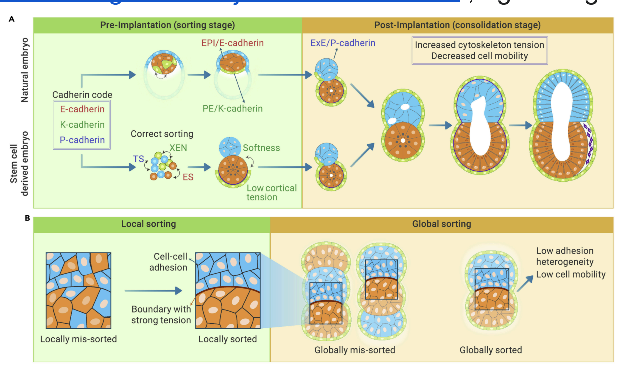
Reference model description (Ver.2 for OpenVT Workshop, Graz, 12th July 2026)

Cell Sorting from Differential Adhesion

SUMMARY:

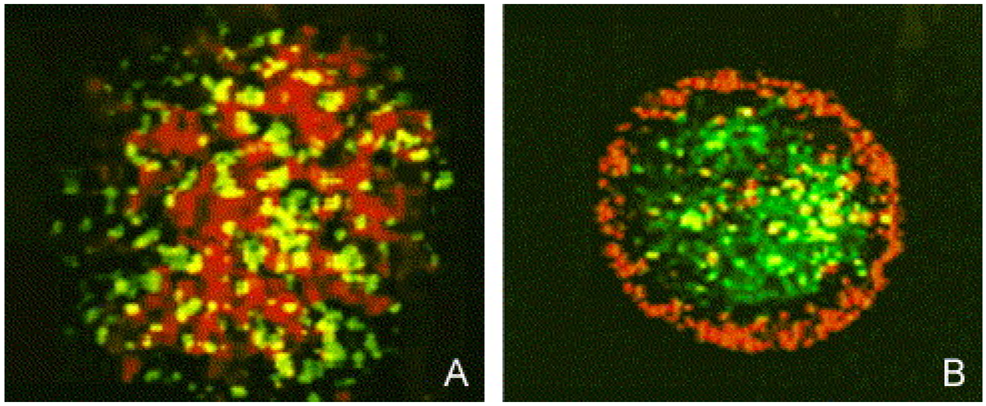
- Given a spatial cell arrangement of 2 cell types in 2D with only volume exclusion and differential cell adhesion as cell-cell interactions, obtain sorting and reduction of heterotypic contact length between cells of different type.
- To allow or speed up neighbor exchange, isolated cells are all equally motile according to a random walk model that has been established previously and is provided as a basis here.
- Explore different parametrizations of the adhesive interaction that yield (A) two compact clumps vs. (B) cell type mixing vs. (C) a single clump engulfed by the other cell type. Quantify and compare time courses of heterotypic contact length in each case through a common analysis pipeline.
- Biological motivation:
  "What did each of us achieve already on day 4 (of our embryonic blastocyst)? We got our mixed Primitive Endoderm (PrE) and Epiblast (Epi) cells sorted apart. Congratulations! The evidence:
  - Hill, M.A. (2026) *Embryology*, left image and watch microscopy movie: https://embryology.med.unsw.edu.au/embryology/index.php?title=Blastocyst_Day_3-6_Movie
  - Bao (2022) *Self-organization principles in stem-cell-derived synthetic embryo models.* The Innovation 4(1), 100366. https://doi.org/10.1016/j.xinn.2022.100366 , right image:





  - Original differential adhesion hypothesis: Steinberg (1962). *On the mechanism of tissue reconstruction by dissociated cells, I. Population kinetics, differential adhesiveness, and the absence of directed migration*. PNAS 48 (9), 1577. https://doi.org/10.1073/pnas.48.9.1577
  - In vitro test: Foty and Steinberg (2005) *The differential adhesion hypothesis: a direct evaluation.* Dev. Biol. 278, 255. https://doi.org/10.1016/j.ydbio.2004.11.012 Fig.4 is basis

for our model with (A) initial and (B) final state:



MODEL REFERENCEs:

- Publications:
  - Graner and Glazier (1992) *Simulation of biological cell sorting using a two-dimensional extended Potts model.* PRL 69, 2013. https://doi.org/10.1103/PhysRevLett.69.2013
  - Osborne et al. (2017) *Comparing individual-based approaches to modelling the self-organization of multicellular tissues.* PLOS Comp Biol 13(2): e1005387. https://doi.org/10.1371/journal.pcbi.1005387 with documentation and Chaste code at https://github.com/Chaste/CellBasedComparison2017/blob/main/test/TestCellSortingLiteratePaper.hpp
  - Recent (poissonianCPM) model for blastocyst sorting: Belousov et al. (2024) PRL 132, 248401. with Quantification of Epi-PrE sorting in ICM of mammalian blastocyst (Fig.3c). https://doi.org/10.1103/PhysRevLett.132.248401)
- Interactive simulation and visualization:
  - Center-based cell sorting in browser (by Randy Heiland): https://rheiland.github.io/eigen_models/cell_sorting_v0.html
  - CPM-based cell sorting in browser with Artistoo (by Inge Wortel): https://artistoo.net/explorables/Explorable-CellSorting.html
  - CPM-based cell sorting with Morpheus (by Leah Edelstein-Keshet): https://identifiers.org/morpheus/M2007

MODEL ASSUMPTIONS:

- Time:
  - Runtime depends on initial condition and parameter regime, aim at observables reaching within 5% of equilibrium (asymptotic values)
- Space:
  - Length is scaled so that results are reported in cell diameters (the diameter of a relaxed cell).

- Initial conditions:
  - Initial spatial domain is rectangular with regular hexagonal packing,
  - 2 domain sizes are used,
    i. one holding 100 cells (for model development) and

ii. another holding 400 cells (for production runs), all cells are initially placed in equilibrium in regular hexagonal lattice.
- Cell type initial configuration
  i. Random checkerboard: cells randomly chosen to be type A or type B 50% exactly of each type (useful to study sorting)
  ii. Block pattern grid of 10x10 or 20x20 cells with all in upper half of type A and all in lower half of type B (useful to study engulfment or mixing)

- Boundary conditions:
  - Results should be independent of the specific type of boundary condition as cells remain in contact with the cluster in the center of the domain. Frameworks may choose what is most convenient with preference for Dirichlet boundary conditions of “medium” state (if “medium” is a valid state in that framework). If needed, domain size could be extended to avoid boundary contact.

BIOLOGICAL PROCESSES:

- Cell-cell volume exclusion and repulsion upon compression (e.g. via potential functions)
- Cell motility according to a simple random walk model,
  - In isolation cells experience a non persistent random walk.
  - Have mean square cell displacement of $1CD^2$ per 100 units of time.
  - (Existing reference model for a (persistent) random walk.
    - https://github.com/tjsego/PersistentCell/blob/start/schemas/models.md
    - Note this is for a persistent random walk but could provide guidance.)
- Cell-cell interaction from adhesive (e.g. cadherin) bonds between cell membranes and change the magnitude of interactions between cells of types A and B
  - Homotypic adhesion strengths between cells of same type are given by
    - γ(A, A)
    - γ(B, B)
  - Heterotypic adhesion strength between cells of different types is
    - γ(A, B)
  - Depending on model formalism, a third (auxiliary cell) type may represent the surrounding “medium” or “void”. Adhesion to “medium” or “void” is considered individually as
    - γ(A, void)
    - γ(B, void)

  - Parameter regimes: (from google)
    - *Sorting*
      - γ(A, B) < γ(A, A) = γ(B, B) **<** γ(A, void) = γ(B, void)
    - (Optional) Engulfment
      - γ(A, A) < γ(A, B) = γ(B, B) **<** γ(A, void) < γ(B, void)
    - (Optional) *Mixing/Checkerboard*
      - γ(A, B) > γ(A, A) = γ(B, B) **=** γ(A, void) = γ(B, void)
- Nothing else.
  - i.e. there are no signals in the microenvironment, so no cell chemotaxis, etc.
  - Cells do not divide nor die.

DATA COLLECTION:

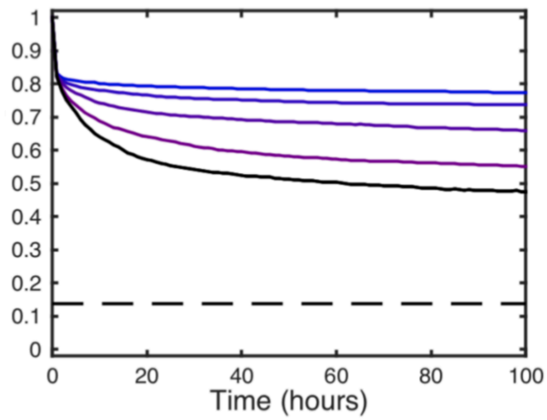
- For **101 time points at even intervals** (t_0=0 t_101=t_end) export the list of cell center locations and cell type for each cell in the format specified below.

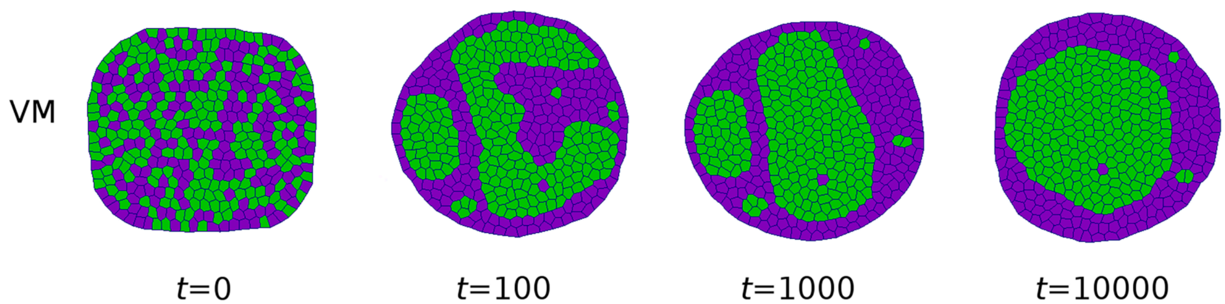
- A common data analysis portal allows to reconstruct cell-cell contact lengths and the total heterotypic contact length for each time point. http://178.105.183.207:3838/OpenVtCellSortingR/
- Data format
    - simID, time, cellID, cellType, x, y ← HEADDER
      ID, t_0, type_1, x_1, y_1
      ID, t_0, type_2, y_2 x, y_2
      …
      ID, t_0, type_N, x_N, y_N
      ID, t_1, type1, x_1, y_1
      ID, t_1, type_2, x_2, y_2
      …
      ID,, t_101, type_N, x_N, y_N
    - Example data for the sorting regime can be found here: https://drive.google.com/file/d/16rEFXieMleE4jABhkaTyLTd6r07LVWc1/view?usp=sharing
    - Note simID

- **The data analysis below assumes that cells can be approximated by spheres of diameter 1 (i.e a typical separation of cell centres is therefore all results need to be scaled so that cells have an approximate diameter of 1.**

DATA ANALYSIS:

- We want to present the heterotypic length (sum of contact lengths between cells of different type, normalised so initial length = 1) over time
- These contact lengths are defined differently in each framework so we provide a consistent calculation based on the Voronio tessellation of the cell centers.
- For consistency here to calculate the heterotypic length we
  1) Build Delaunay triangulation of real cell centers.
  2) Remove Delaunay edges longer than a threshold (default 2.0). This removes connections from cells that are too far apart.
  3) Add ghost nodes at distance 1.0 outside each boundary edge:
     - Two ghosts per boundary edge, at 10% and 90% along the edge.
  4) Compute Voronoi tessellation of real + ghost nodes.
  5) Sum Voronoi edge lengths between heterogeneous real-cell pairs only.
- The portal will centrally provide summary statistics like the heterotypic length time course for each run and plot end cell patterns.
- Here is an example of heterotypic length (normalised) for a vertex model simulation form Osborne

- 



EXAMPLE GUIDING PARAMETRIZATION:

- Individual model formalisms require their own set of parameters to see the behaviour detailed above and we deliberately don't include any here. However Osborne et al. (2017), provides example parameters used for sorting See Tables 1 and 2: Note: these are a starting point and won't work for our examples.
    - https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1005387
- Another good source of parameters is Galzier and Graner https://journals.aps.org/pre/pdf/10.1103/PhysRevE.47.2128